

Introduction to Haematology

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Introduction to Paediatric Haematology / Investigations

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Outline

Introduction · Overview of blood · Functions of blood · Sub-classification of paediatric haematology
· Investigations

Introduction

Haematology is the branch of medicine concerned with the **study, diagnosis, treatment and prevention of blood-related diseases**.

- **Blood is one of the largest organs in the body.**
- It contains **plasma (~60% of total blood volume)** and **cells (~40%)**. This percentage **changes with age** – in a **preterm baby, plasma can be up to 70%**.

Overview of blood

Know the different **cell lines** of blood and how they are formed – **erythrocytes (RBCs), leucocytes (WBCs) and platelets**. By number, **RBCs are most abundant, followed by platelets, and lastly WBCs**, regulated by a **feedback mechanism**.

Haematopoiesis (blood formation) occurs at different anatomic sites during life:

- **Yolk sac** – 2nd-3rd week of gestation, diminishing by the 10th week.
- **Liver** – 6th week up to the 30th week.
- **Bone marrow** – from the 20th week and continuing till (and after) birth.

Extramedullary haematopoiesis – a situation where the **liver and spleen** are called upon to make blood in times of **stress and increased demand** for more cells.

Functions of blood

- **RBCs** – oxygen transport; also act as a **buffer** (high carbonic anhydrase).
- **Granulocytes (BEN – Basophils, Eosinophils, Neutrophils)** – a subgroup of WBCs that kill microorganisms; they are the **first line of defence**.
- **Lymphocytes** – produce **antibodies** and mediate **cell-mediated immunity (CMI)**. **B cells** -> antibodies; **T cells** -> **cellular immunity**.
- **Platelets** – **haemostasis** (blood clotting).

Sub-classification of paediatric haematology

1. **Diseases affecting blood production:**
 - **Decreased production** – e.g. anaemias, leucopenias, thrombocytopenias.
 - **Increased production** – e.g. polycythaemias, leucocytosis, thrombocytosis.
2. **Abnormal haemoglobins** – sickle cell anaemia, thalassaemia.
3. **Coagulation problems** – e.g. DIC (and tropical splenomegaly syndrome / hyper-immune splenomegaly – the spleen in TSS).
4. **Enzyme defects** – e.g. G6PD deficiency.

Investigations

- **Hb or PCV.**
- **FBC** – reticulocyte count, Hb, total and differential WBC, RBC count, platelet count.
- **Peripheral blood film** – blast cells, RBC morphology.
- **Genotype.**
- **Bone marrow aspiration** – cytology.
- **Bone marrow biopsy** – histology.

References

Lecture material of Prof I.F. Ogbonna (500-level Paediatrics, Abia State University); Nelson Textbook of Paediatrics.